

RSG-HRGV: theory properties and directly matched evidence

The statements below characterize expert routing; they do not assert a universal classification gain.

Theorem B.1 Routing regret upper bound

For $a=p_d(y|x)$, $b=p_m(y|x)$, $g_o=I[a>=b]$, and $a,b>=epsilon$:
 $-\log p_g(y|x) + \log \max(a,b) \leq |g-g_o| |a-b| / \epsilon$.

Theorem B.2 Soft-oracle approximation

$\Delta = l_m - l_d$, $g^* = \text{sigmoid}(\Delta/T_r)$. The deviation from hard optimal routing is bounded by $\exp(-|\Delta|/T_r)$; small gaps retain uncertainty instead of forcing a switch.

Theorem B.3 Local gradient isolation

$L_{reg} = w \text{ BCE}(g, g^*)$, $w = \tanh(|\Delta|/T_w)$. Applying stop-gradient to the regret branch gives $dL_{reg}/d\theta_e = 0$ for backbone and expert parameters through this branch.

$$p_g = g p_d + (1-g)p_m, \quad g^* = \text{sigmoid}((l_m - l_d)/T_r), \quad w = \tanh(|l_m - l_d|/T_w)$$

Fixed test

-1.77 pp [-2.86 pp, -0.69 pp]

-3.53 pp [-4.75 pp, -2.17 pp]

Photographer holdout

Both 95% bootstrap intervals remain below zero.
This confirms reduced predefined routing regret, not classification superiority.

classification superiority is not claimed

Difference in mean routing regret (percentage points; lower is better)

Predefined routing-regret effect (RSG-HRGV minus HRGV)